

Online Shoppers Purchasing Intention

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Topic Outline

A

The Business Problem

B

Understanding Customer Behavior

C

Key Insights

D

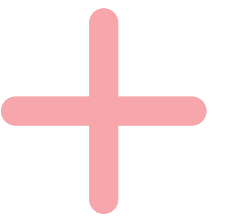
Purchase Prediction



Introduction

Online shopping creates valuable customer behavior data.

**But one question remains:
What makes a visitor decide to buy?**



The Business Problem

The Big Question: Where Are We Losing Shoppers?

Conversion Imbalance

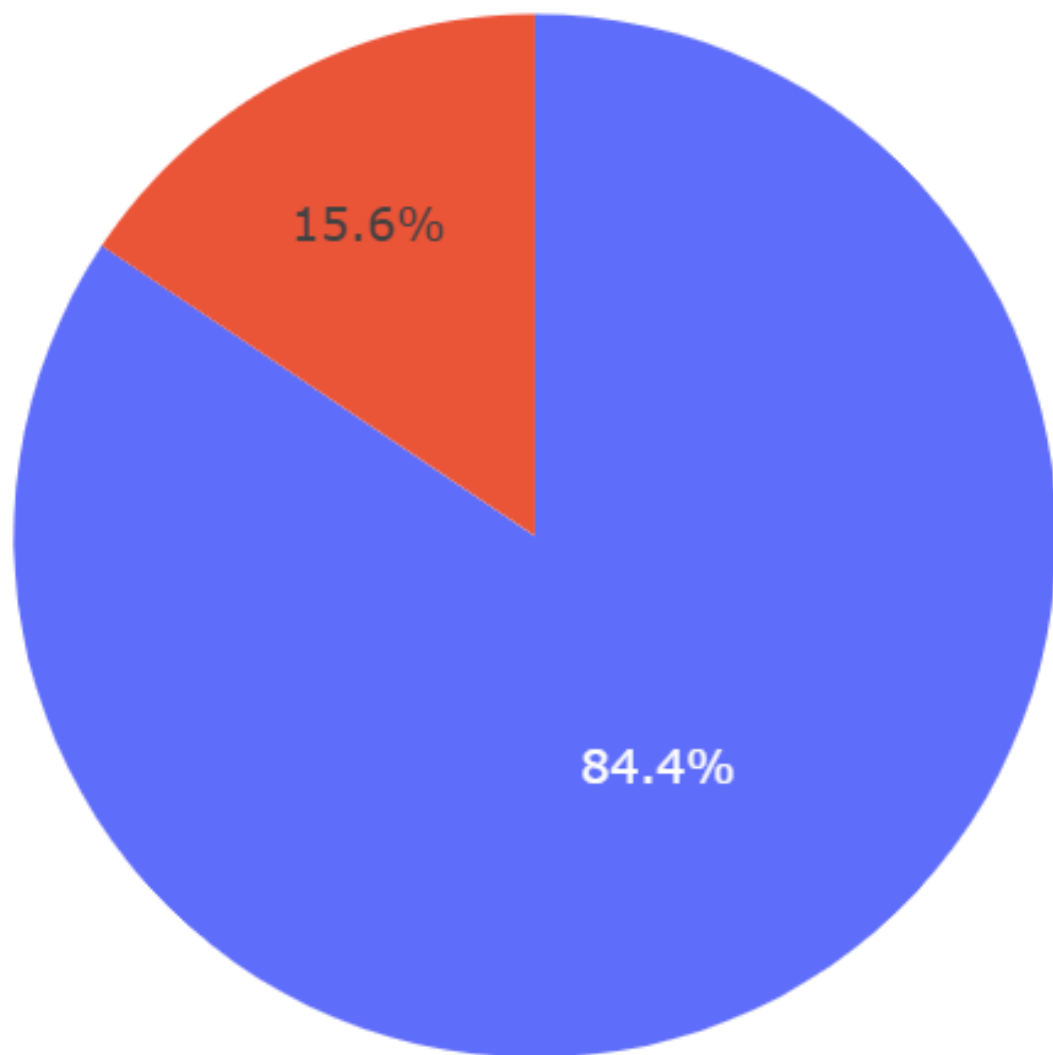
Out of 12,330 sessions, only 15.5% resulted in a purchase, while 84.5% ended without a purchase.

The Challenge

A large number of visitors leave without completing a purchase.

Key Question

How can we identify high-intent shoppers and turn more browsers into buyers?



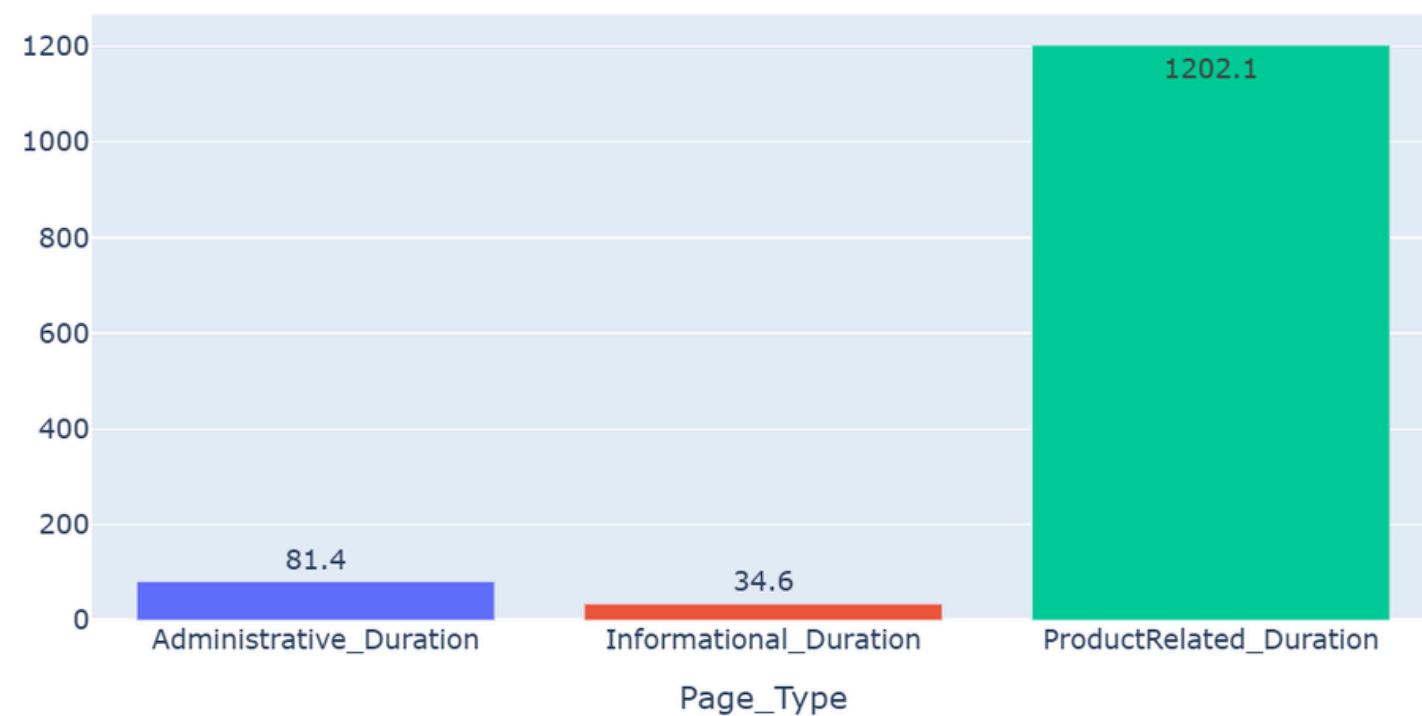
Where Do Shoppers Lose Interest?

Bounce Rate :2.2%

Exit Rate: 4.3%

Key Insight:Non-purchasing sessions show higher bounce and exit rates.

This suggests that some shoppers lose interest before completing a purchase.



Physical

PageValue

What is PageValue?

Measures the value of pages visited before transaction.

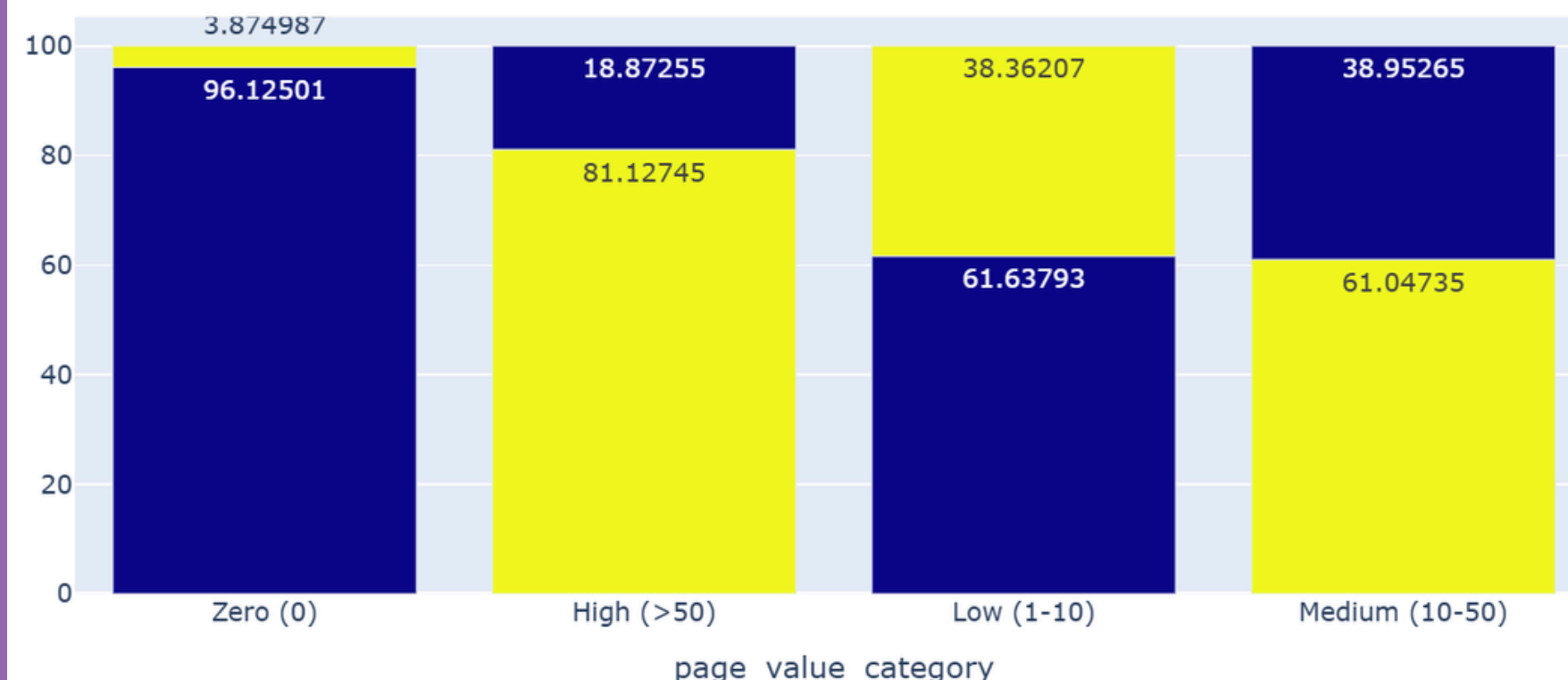
The Contrast

Purchased : PageValue > 25

Not Purchased : PageValue $\approx$ 0

Key Insight :PageValue strongly distinguishes purchasing sessions from non-purchasing sessions.

Revenue Distribution by Page Value Category



Who Converts Better: New or Returning Visitors?

Returning Visitors — ~85% of sessions

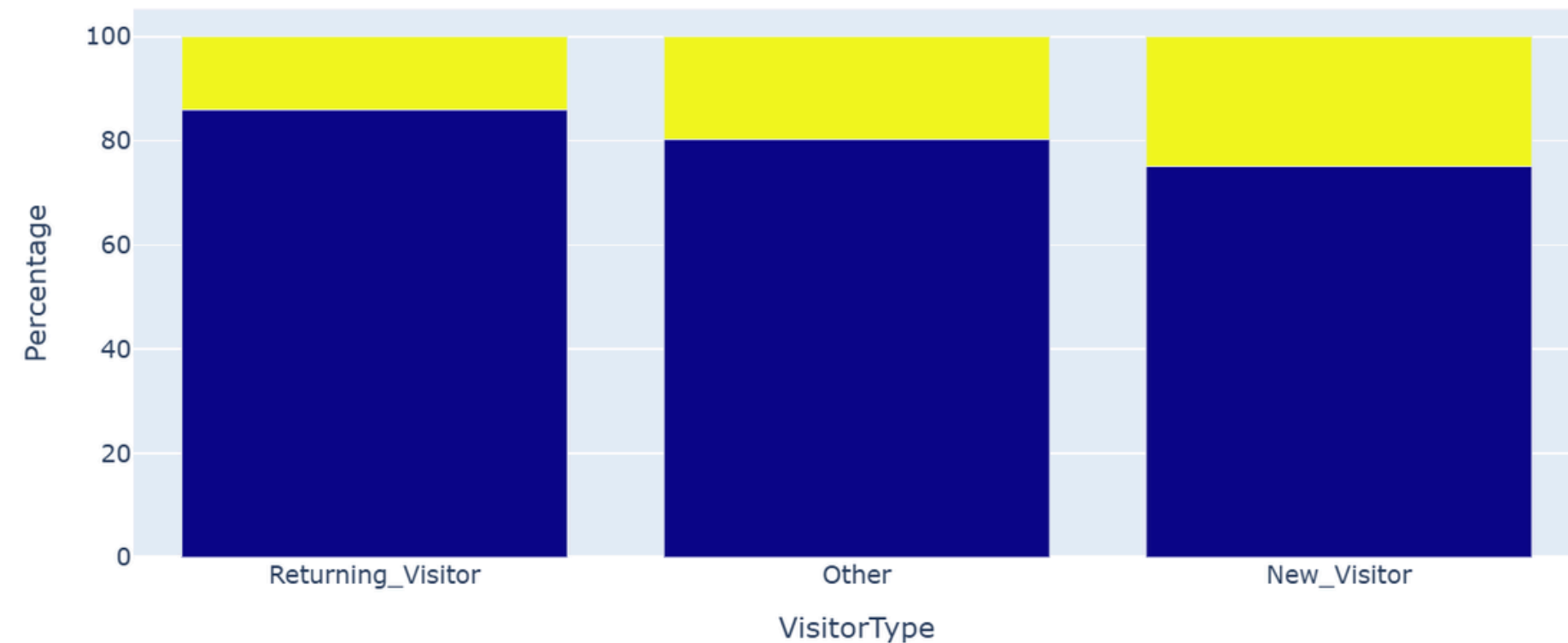
High traffic volume :~14–15% conversion

New Visitors — ~15% of sessions

Lower traffic volume : ~25% conversion

Key Takeaway: New visitors convert more efficiently, while returning visitors drive higher traffic volume.

Revenue Distribution by Visitor Type



Predicting Purchasing Intent: Machine Learning Pipeline

Classification Task:

Predict Buyer vs. Non-Buyer using Revenue.

Data Preparation:

Encode categorical features and transform/scale numerical features.

Class Imbalance:

Address the 15.5% vs. 84.5% imbalance to avoid bias toward non-buyers.



Finding the Winning Algorithm

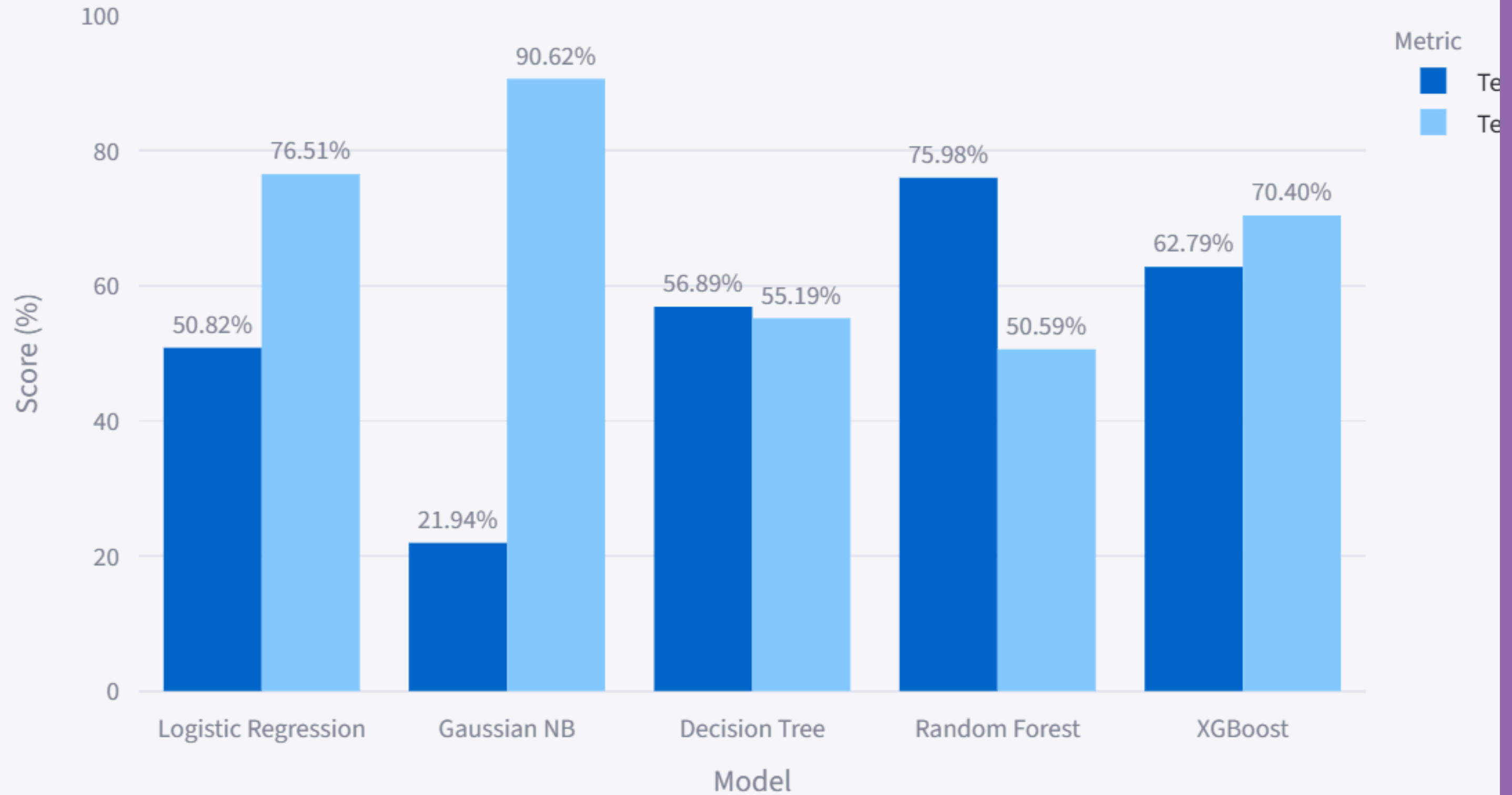
Model Comparison

Model	Test Precision	Test Recall
Logistic Regression	50.82%	76.51%
Gaussian Naive Bayes	21.94%	90.62%
Decision Tree	56.89%	55.19%
XGBoost ★	62.79%	70.4%

Key Takeaway: XGBoost achieved the best balance between Precision and Recall, making it the final model.



Test Precision vs Recall



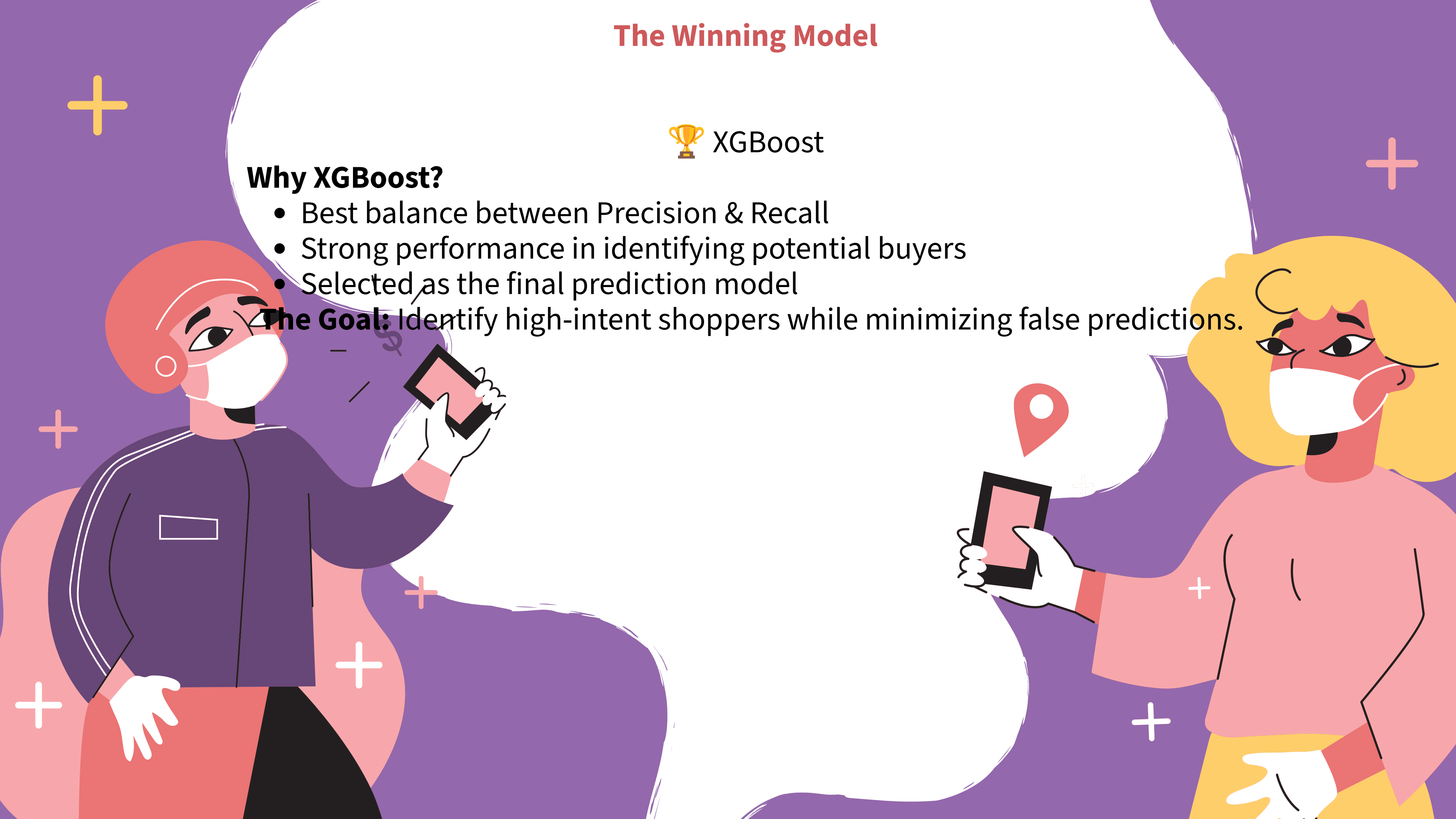
The Winning Model

🏆 XGBoost

Why XGBoost?

- Best balance between Precision & Recall
- Strong performance in identifying potential buyers
- Selected as the final prediction model

The Goal: Identify high-intent shoppers while minimizing false predictions.



Optimizing XGBoost

Best Configuration

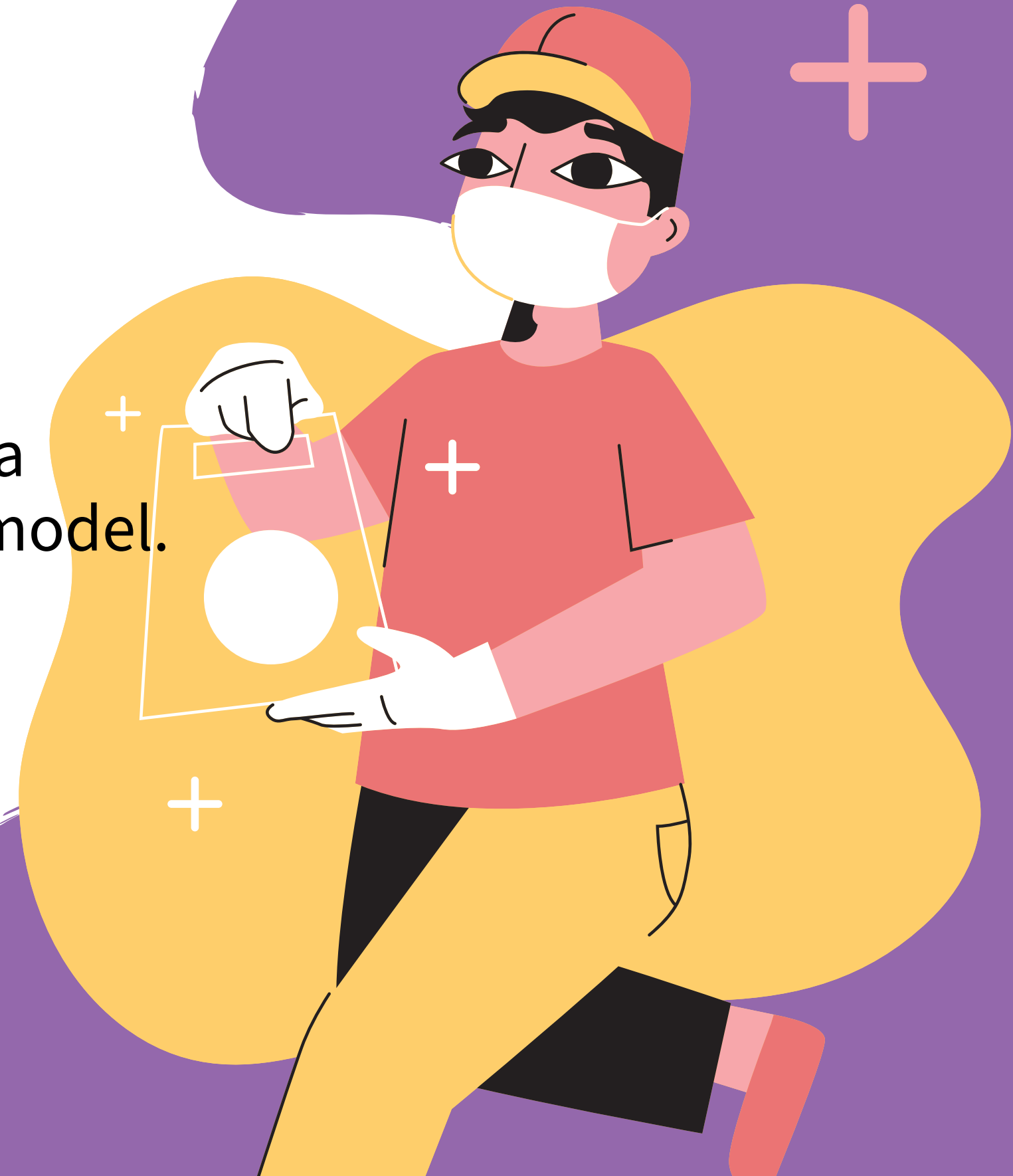
Max Depth: 5

Number of Estimators: 50

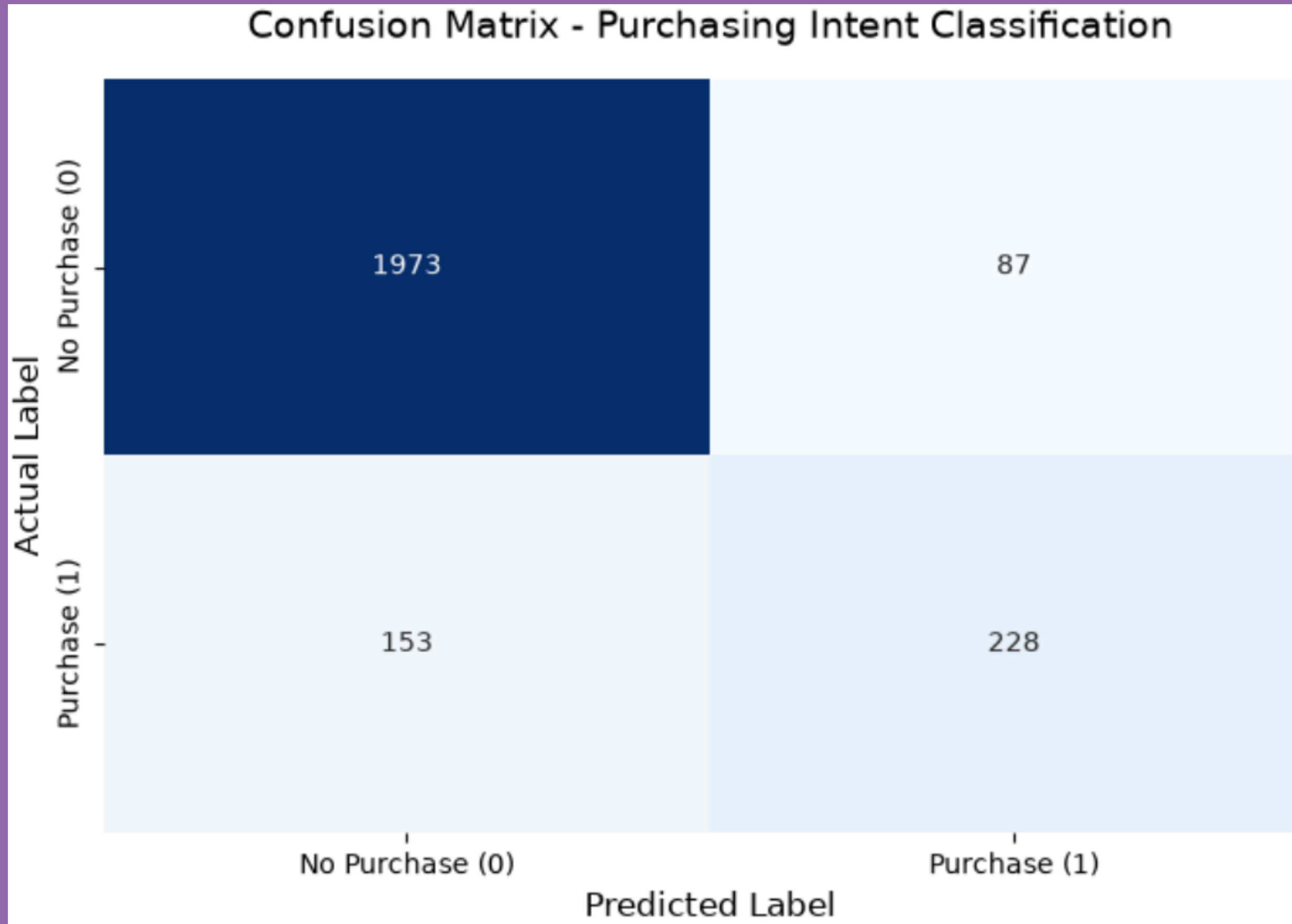
Cross-Validation Performance

- Best Mean Test Score :78.47%
- Best Mean Train Score :96.18%

Key Takeaway :Hyperparameter tuning identified a more suitable XGBoost configuration for the final model.



Final Model – XGBoost



Recommendations

1. Optimize High-PageValue Paths

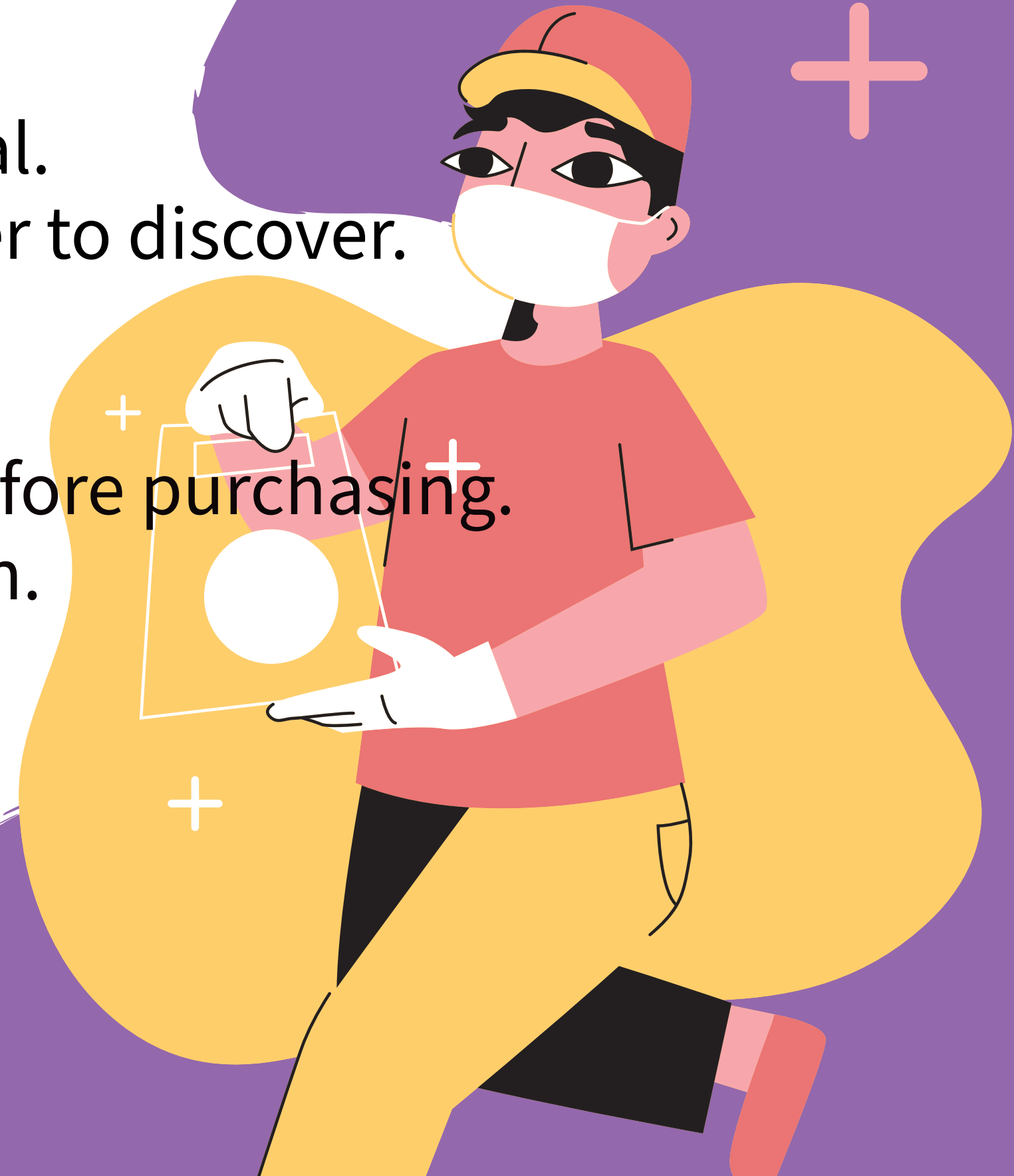
Insight: PageValue is a strong purchase signal.

Action: Make high-value product pages easier to discover.

2. Reduce Drop-Offs

Insight: Engaged shoppers may still leave before purchasing.

Action: Simplify checkout and reduce friction.



Recommendations

.3. Re-engage Returning Visitors

Insight: Returning visitors drive most of the traffic but convert less efficiently.

Action: Use personalized recommendations and re-engagement campaigns.

4. Prepare for Peak Months

Insight: Shopping activity peaks in May & November.

Action: Strengthen marketing and website capacity during peak periods.





Thank you for listening!

Don't hesitate to ask any questions!

